



ALL INDIA INSTITUTE OF MEDICAL SCIENCES
BHUBANESWAR, ODISHA
DEPARTMENT OF PHYSIOLOGY
ANNOUNCES



PHYSICON 2025

The XXXVI Annual Conference of The
Physiological Society of India

“Innovation in Physiology: From Cell to Systems”

Pre-Conference workshop

20th November, 2025

Conference :

**21st – 23rd November,
2025**

Organized by :

**Department of
Physiology,
AIIMS Bhubaneswar**

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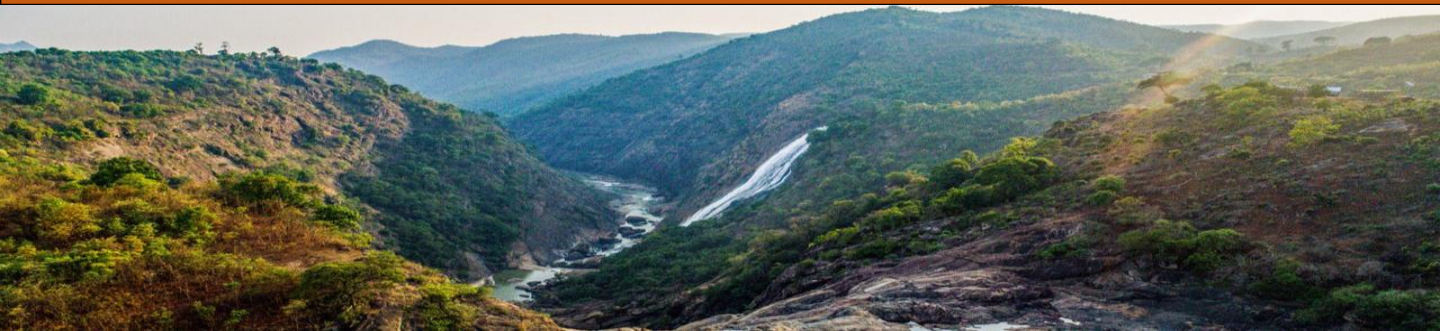
SCIENTIFIC PROGRAM OF PHYSICON 2025

Time	DAY 1 (21.11.2025, Friday)		
8:00-9:00 AM	Registration & Breakfast		
9:00-10:30 AM	Inauguration		
10:30-10:45 AM	Tea Break		
10:45-12:00 PM	Key Note Address 1 Main Auditorium	Prof. A.K. Mahapatra President, AIIMS Bhopal Founding Director AIIMS, BBSR	
	Key Note Address 2 Main Auditorium	Prof. Bhanu Pratap Jena George E. Palade University, Professor and Distinguished Professor, Wayne State University School of Medicine <i>The Porosome Secretory Nanomachine: Discovery to Therapy</i>	
	Key Note Address 3 Main Auditorium	Prof. T R Raju Director, Research, Sankara Academy of Vision, Bengaluru <i>Unravelling the Pathophysiology of ALS- The Motor Neuron Disease: From Cells to Systems</i>	
12:00-1:30 PM	<u>Main Auditorium</u> Chairpersons Prof. S C Mahapatra, Ex-Dean (A), Ex-HOD, Physiology, AIIMS, BBSR Prof. G K Pal Senior Scale Prof JIPMER <u>Plenary Speaker</u> Dr Rajen Halдар Calcutta University, West Bengal <i>Erythrocyte in PCOS before and after therapeutic interventions.</i>	<u>Lecture Theatre-2</u> Chairpersons Prof. Swagata Tripathy, AIIMS, BBSR Prof. Seema Singh, ERA’S Medical College, Lucknow <u>Plenary Speaker</u> Dr. Ravindra P.N NIMHANS, Bengaluru <i>Neuro-Phenomenological landscape of sleep</i>	<u>Lecture Theatre-3</u> Chairpersons Prof. Sanjit Dey, University of Calcutta, Kolkata Prof. Snigdha Prava Mishra, IMS & SUM Hospital Campus II
	Symposium 1 <i>Three Pillars of Physiology: Innovation, Education & Research</i> 1. Prof. Prasunpriya Nayak, AIIMS Jodhpur <i>Revisiting Research - the Forgotten Core of Physiology</i> 2. Prof. Abhinav Dixit, AIIMS, Jodhpur <i>Innovations: Dawn of a New Era for Physiologists</i> 3. Prof. Bharati Mehta, AIIMS, Jodhpur <i>Physiology Education: Teaching to Transform, Assessing to Advance</i> 4. Prof. Sarmishtha Ghosh, Freelance physiologist- health professions educator/mentor <i>AI Powered Physiology Teaching</i>	Symposium 2 <i>Is Sleep the Architect of Health?</i> 1. Prof. Arpita Priyadarshini, Fellow AASM, DRIEMS Medical College & Hospital, Cuttack, Odisha <i>An Era of Ruined Sleep & its Outcome</i> 2. Dr. Roopam Panda, SCBMCH, Cuttack <i>Creation of Sleep</i> 3. Dr. Meena Mirdha, AIIMS, Bhatinda <i>Sleep Architecture a Marvel: Part 1 NREM</i> 4. Dr. Barsha Baishali Parida KIMS ,BBSR <i>Sleep Architecture a Marvel : Part 2 REM</i>	<u>Faculty Oral Presentation (FOP)</u> FOP1 - FOP8
1:30-2:15 PM	Lunch Break		

Time	DAY 1 (21.11.2025, Friday)			Scientific Program
2:15 - 3:45 PM	<u>Main Auditorium</u> <u>Oration Awards</u> 1. Prof. S. C. Mahalanobis Memorial Oration: Dr. Subhrajit Saha, University of Kansas Medical Centre, USA 2. Sri. Ramendra Sundar Sinha Memorial Oration: Dr. Varun Malhotra, AIIMS, Bhopal 3. Prof. P. B. Sen Memorial Oration: Dr. Yasmin Ahmad, DIPAS, Delhi 4. Prof. S. R. Maitra Memorial Oration: Dr. Amit Bandyopadhyay, University of Calcutta, Kolkata	<u>Lecture Theatre-2</u> <u>Chairpersons</u> Prof. B. R. Mishra, Dept of Psychiatry, AIIMS, BBSR Prof. Sanjeev Bhoi, Dept of Neurology, AIIMS, BBSR <u>Plenary Speakers</u> Prof TN Sathyaprabha NIMHANS, Bengaluru <i>ALS Research at NIMHANS - Bed to Bench side</i> Prof. Kaviraja Udupa NIMHANS, Bengaluru <i>Next-Generation Neuromodulation: Personalized Therapies for Neurological and Psychiatric Disorders</i>	<u>Lecture Theatre-3</u> <u>Chairpersons</u> Prof. Abhinav Dixit, AIIMS, Jodhpur Prof. R N Sahu, Dept of Neurosurgery, AIIMS, BBSR <u>Plenary Speaker</u> Prof. BSS Rao NIMHANS, Bengaluru <i>Novel Strategies of Treat Temporal Lobe Epilepsy-induced Cognitive Deficits and Abnormal Synaptic Plasticity</i>	
		Symposium 3 <i>Next-generation Neuro modulation: Personalized Therapies for Neurological & Psychiatric Disorders</i> 1. Dr Anant Kumar, AIIMS Vijaypur, Jammu <i>Update on Neurophysiology and TMS based Neuro modulation in Chronic Form of Primary Headache</i> 2. Dr. Ashish Aravind, AIIMS, Gorakhpur <i>Neurophysiology of Primary Headache</i> 3. Dr. Bhawna Mattoo, ESICMCH, Faridabad <i>TMS-based Neuro modulation in Chronic Tension-type Headache</i>	Symposium 4 <i>Cognition & the Brain: A Neuro-physiological Exploration Using EEG</i> 1. Dr. Megha Agrawal, AIIMS, Jammu <i>Automated Detection and Classification Of Seizure Types From Interictal EEG Signals.</i> 2. Dr. Amal Mondal, JNU, Delhi <i>MST1 Selective Inhibitor Xmu-mp-1 Ameliorates Sporadic Alzheimer's Disease in a Rat Model by Modulating Hippo-Wnt Signaling Crosstalk</i> 3. Prof. Pranati Nanda, KIMS,BBSR <i>Cognition in Autism Spectrum Disorder</i>	
3:45- 4:00 PM Tea Break				
4:00-6:00 PM	<u>Main Auditorium</u> <u>Oration Awards</u> 1. Prof. A. K. Mukherjee Memorial Oration: Dr. D Ghosh, DIPAS, Delhi 2. Smt. Shakuntala Das Gupta Memorial Oration: Dr. Krishnendu Manna, University of Kalyani, WB 3. Prof. P.B. Sen Memorial Research Prize: Dr. Rahul Debnath, down town University, Assam 4. Prof. B. B. Sarkar Memorial Research Prize: Dr. Arnab Das, RKMVERI, West Bengal. 5. Prof. B. N. Koley Memorial Oration: Dr. Manpreet Kaur, VMMC & SJH, Delhi	<u>Lecture Theatre-1</u> <u>Chairpersons</u> Prof. Soumen Manna, VMMC, Delhi Prof. Madhuchhanda Pattnaik, DD Medical College, Keonjhar <u>Faculty Oral Presentation (FOP)</u> FOP9 – FOP11 <u>Students Oral Presentation (SOP)</u> SOP1 – SOP8	<u>Lecture Theatre-2</u> <u>Students Oral Presentation (SOP)</u> SOP9 – SOP20	
5:00-6:00 PM	<u>Main Auditorium Foyer</u> Poster Presentation (FPP1-FPP9 and SPP1-SPP19)			
7:00 -8:30 PM	Cultural Programme			
8:30 PM onwards	Gala Dinner			

Time	DAY 2 (22.11.2025, Saturday)		Scientific Program
8:00-9:00 AM	Registration & Breakfast		
9:00-10:00 AM	Key Note Address 4 <u>Main Auditorium</u>	Prof. G K Pal Sr Scale Prof, JIPMER, Puducherry <i>Experimental Physiology Research: The Needs & Challenges of it's Revival, Survival & Upheaval in the New era of Emerging Clinical & Interventional Physiology</i>	
	Key Note Address 5 <u>Main Auditorium</u>	Dr. Madhusudan S Pal, Scientist G. DIPAS, Delhi <i>Physiological and Ergonomics Challenges: Infantry Combat Vehicle Design</i>	
10:00 – 11:30 AM	<u>Main Auditorium</u> <u>Chairpersons</u> Prof. Manisha Kar, AIIMS, BBSR Prof. Bharati Mehta, AIIMS, Jodhpur <u>Plenary Speaker</u> Prof. Sovan Bagchi, Gulf Medical University, UAE <i>Impact of Microglia in Neurodevelopmental Disorders</i> Prof. Iqbal Alam, HIMSAR, Delhi <i>Therapeutic Potential of Asparagus racemosus in Preeclampsia: Modulation of Sterile Inflammation and Angiogenic Pathways</i>	<u>Lecture Theatre-2</u> <u>Chairpersons</u> Dr. J. Dandapat, Utkal University, BBSR Dr. Rupesh Dash, ILS, BBSR <u>Plenary Speaker</u> Prof. Chandan Goswami, NISER,BBSR <i>TRP channels in sub-cellular organelle structure and functions:Importance in lysosomal biology</i> Dr Kalpana Barhwal, AIIMS, BBSR <i>Mitophagy-Ferroptosis Crosstalk in Glioma: A Double-Edged Sword in Tumor Progression and Therapy</i>	
	Symposium 5 <i>Advances in Genomic and Cellular Physiology</i> Dr. Jeevitha Poorasamy, AIIMS, Jammu Prof Dwaipayan Bharadwaj, SNU, Kolkata Prof. Vimlesh Kumar, IISER, Bhopal	Symposium 6 <i>Enforced Mitophagy: A Therapeutic Game Changer</i> Dr. Sunil Kumar Jena, VIMSAR, Burla Dr. Manju Agrawala, SRM MCH, Kalahandi Dr. Sipra Dash, DD MCH, Keonjhar	
11:30 11:45 AM	Tea Break		
11:45-2 PM	<u>Main Auditorium</u> Oration Awards Dr. Chitralekha Mukherjee Prof. K.Raghothama Rao Dr. Siddhartha Debnath Prof. B. B. Sarkar	<u>Lecture Theatre-3</u> <u>Chairpersons</u> Dr. Pruthviraj Sethi, Dept of OBGY, AIIMS, BBSR Dr Rajalakshmi R, JSS Medical College, Mysuru <u>Plenary Speakers</u> Dr. Guruprasad Kalthur Dean (R), SMCRI, Kanyakumari Dr Rupasri Ain, IICB, Kolkata	<u>Lecture Theatre-4</u> <u>Invited Lectures</u> Dr. Snehashish Bhuniya, Saifai,UP Prof. Amlan Das, TARGU, Assam Dr. Soumendra Darbar, Kolkata Dr. Rituparna Sinha Roy, IISER,Kolkata Dr. Anjana Ray, Bangabasi Evening College Dr D P Dolai, MMU, West Bengal Dr Smriti Ratan Tripathy,BGC, West Bengal Dr. Debasis maiti, Tripura University Dr. Amit Roy, INSA young Scientist, S.P.Pune University Dr. Manjusha Dixit, NISER, Bhubaneswar Dr. Madhumita Debnath, KBC College, Howrah
	Dr. Sankar Kumar Dey, SBSS Mahavidyalaya, West Bengal Prof. Bharti Bhandari Rathor, GIMS, Greater Noida Dr. Arnab Banerjee, Serampore College, West Bengal Prof. Jayanti Mishra, AIIMS, BBSR	Symposium 7 <i>Molecular Insights Into Reproductive,Endocrine and Gastrointestinal Physiology</i> Dr. Barnali Basu Ray, University of Kolkata Dr Pratip Chakraborty, Cradle Fertility Center, Kolkata Prof. Sanjit Dey,University Of Calcutta Dr. Asima Bhattacharya, NISER,BBSR Dr. Sudipta Pal, Tripura University	Dr. Snehashish Bhuniya, Saifai,UP Prof. Amlan Das, TARGU, Assam Dr. Soumendra Darbar, Kolkata Dr. Rituparna Sinha Roy, IISER,Kolkata Dr. Anjana Ray, Bangabasi Evening College Dr D P Dolai, MMU, West Bengal Dr Smriti Ratan Tripathy,BGC, West Bengal Dr. Debasis maiti, Tripura University Dr. Amit Roy, INSA young Scientist, S.P.Pune University Dr. Manjusha Dixit, NISER, Bhubaneswar Dr. Madhumita Debnath, KBC College, Howrah
2 - 2:30 PM	Lunch Break		

Time	DAY 2 (22.11.2025, Saturday)			Scientific Program
2:30 - 4:15 PM	<u>Main Auditorium</u> <u>Chairpersons</u> Prof. Ashutosh Biswas, Director, AIIMS,BBSR Prof. Barathram, IIT, BBSR <u>Plenary Speaker</u> Dr. K K Deepak CBME IIT Delhi <i>Innovations for Physiology & Medicine: Whose responsibility?</i>	<u>Lecture Theatre-1</u> <u>Chairpersons</u> Dr Sudipta Pal, Tripura University Prof. Ekta Khandelwal, AIIMS, Raipur <u>Plenary Speaker</u> Prof. Meenakshi Sinha AIIMS Raipur <i>Mental Health in Primitive Communities : Neurophysiological & Genetic Challenges</i>	<u>Lecture Theatre-2</u> <u>Chairpersons</u> Prof. Iqbal Alam, HIMSAR, Delhi Prof. Samata Padaki, K.H Patil Institute of Medical Sciences, Karnataka <u>Faculty Oral Presentation (FOP)</u> FOP12 -FOP21	
	Symposium 8 <i>Application of Artificial Intelligence in Physiological Sciences</i> 1. Dr. Anupkumar D. Dhanvijay, AIIMS, Deoghar <i>AI in Physiology Education: From Case Vignettes to MCQ Item Analysis</i> 2. Dr. Mohammed Jaffer Pinjar, AIIMS, Deoghar <i>AI as a Research Companion: Writing, Publishing Peer Review in Physiology</i> 3. Dr. Abhimanyu Ganguly, AIIMS, Deoghar <i>AI at the Intersection of Physiology & Healthcare Research</i> 4. Dr. Prathamesh Kamble, AIIMS, Nagpur <i>Development of brain-computer interface system</i>	Symposium 9 <i>Impact of Air Pollution: Bridging b/w Engineering & Medicine</i> 1. Prof. Asmita Patil AIIMS, Delhi 2. Dr. Debabrata Dasgupta IIT, Delhi 3. Ms. Swarnabha Sarkar AIIMS, Delhi <i>Impact of Pollution on Reproductive System</i> 4. Dr. Revand R. AIIMS, New Delhi <i>Impact of Pollution on Cardio-respiratory Functions</i> 5. Dr. Aditya Dontham GMC, Vizianagaram <i>Impact of Pollution on Metabolic Functions</i> 6. Ms. Pushpanjali, AIIMS, Delhi		
4:15-4:30 PM				



Time	DAY 3 (23.11.2025, Sunday)			Scientific Program
8:00-9:00 AM	Registration & Breakfast			
9:00-10:00 AM	Key Note Address 6 <u>Main Auditorium</u>	Prof. Amar K Chandra UGC Emeritus Fellow, University of Kolkata <i>Structural & Functional Transformation of Thyroid cells through cyanogenic plant foods: An innovative study from cell to system</i>		
	Key Note Address 7 <u>Main Auditorium</u>	Prof. Nihar Ranjan Nayak University of Missouri, Kansas City, USA <i>Placental sFlt-1: A Double-edged Sword in Maternal Vascular Adaptations to Pregnancy</i>		
10:00 – 11:15 AM	<u>Lecture Theatre-1</u> Chairpersons Dr K P Singh, DRDO, Delhi Dr Sunil Hota, DIPAS Delhi Symposium 10 <i>Physiology in Military Platforms</i> 1. Dr. Tirthankar Chatterjee, DIPAS, Delhi <i>Physiological Aspects in Closed Environment Platforms</i> 2. Dr K V Mani DIPAS, Delhi <i>Man Made Environments and Physiological Implications</i> 3. Dr Krishna Kishore DIPAS, Delhi <i>Military Physiology in Clothing and Protective Gears</i>	<u>Lecture Theatre-2</u> Chairpersons Dr Smritiratan Tripathy, Berhumpur Girls College, West Bengal Dr Kishore K Behra, AIIMS, BBSR <u>Plenary Speakers</u> Dr. Amit Krishna De, Sister Nibedita University, Kolkata <i>Anti-inflammatory, analgesic, anti-microbial and anti-oxidant compounds from Indian Spices</i> Dr Amit Bandyopadhyay, University of Calcutta, Kolkata <i>Ergogenic Efficacy of Traditional Medicines</i> Symposium 11 <i>Establishment of an Obesity Evaluation Clinic and Updates on Obesity Management Strategies</i> 1. Dr K. Sarvottam, AIIMS, Gorakhpur Dr. Ritesh Kumar Netam, AIIMS, Delhi <i>The Pathophysiology of Obesity and Practical Aspects of Establishing the Obesity Evaluation Clinic</i> 2. Dr. Samir Kumar Singh, IMS-BHU <i>Establishment of an obesity evaluation clinic & updates on obesity management strategies</i> 3. Dr. Arani Das, AIIMS, Gorakhpur <i>Multidisciplinary Approach to Obesity Management</i> 4. Dr. Ashish Arvind, AIIMS, Gorakhpur <i>Innovative Strategies and Technologies in Obesity Management</i>	<u>Main Auditorium</u> Students Oral Presentations for Awards	
	11:15-11:30 AM	Tea Break		



Time	DAY 3 (23.11.2025)			Scientific Program
11:30-1:00 PM	<u>Lecture Theatre-1</u> Chairpersons Dr K C Mathur Ex-HOD, SPMC, Bikaner Dr Utpal Dutta, Ex-DMET, West Bengal <u>Plenary Speaker</u> Prof. Palok Aich NISER, BBSR <i>Metabolome–Microbiome Crosstalk: Towards Early Diagnosis and Precision Interventions in Metabolic and Inflammatory Disorders</i>	<u>Lecture Theatre-2</u> Chairpersons Dr M Taranikanti, AIIMS Bibinagar Dr. Anish Singhal AIIMS, Bibinagar <u>Plenary Speaker</u> Prof. Ramanjan Sinha, AIIMS Raipur <i>Cognitive deficits in Sickle Cell Disease : Electrochemical evidence of a therapeutic approach</i>	<u>Lecture Theatre-3</u> Chairpersons Prof. Kanishka Das, Dept Pediatric Surgery, AIIMS, BBSR Prof. Sonu Subba, Dept CMFM, AIIMS, BBSR <u>Plenary Speaker</u> Dr. Minal Patel <i>PMC, Karamsad, Gujrat</i> <i>Integrating Near-Peer Teaching in Team-Based Learning: A Novel Approach to Teaching in Physiology</i>	
	Symposium 12 <i>Sarcopenia in Metabolic Syndrome</i> 1. Prof. Dhanalakshmi Y JIPMER Pondicherry <i>Metabolic Syndrome: A Physiologist's Perspective</i> 2. Prof. Velkumary S JIPMER Pondicherry <i>Clinical Determination of Sarcopenia</i> 3. Prof. Pravati Pal JIPMER Pondicherry <i>Mechanisms Linking Metabolic Syndrome & Sarcopenia</i> 4. Dr. Kavitha N JIPMER Pondicherry <i>Sarcopenia & Metabolic Syndrome in Anxiety Disorders</i> 5. Dr Saranya K JIPMER Pondicherry <i>Frailty Status & Cardiovascular Autonomic Dysfunction in Type 2 Diabetes Mellitus</i>	Symposium 13 <i>Breakthrough in Nanomedicine & Nanotherapeutics</i> 1. Dr. Maya Nandkumar SCTIMST,Trivendrum <i>“Nanomedicine” titled “In vitro Systems to Study cell material intractions</i> 2. Dr Jiban Jyoti Panda, INST, Mohali <i>Brain targeting neuronanomedicine</i> 3. Dr Karunakar Kar JNU, Delhi <i>Targeting Amyloidosis Through Nanomedicine Based Approach</i> 4. Dr. Suman Das, AIIMS, Delhi <i>Epilepsy & EEG Feature Analysis for Detecting the Preictal Phase of a Seizure Event</i>	Symposium 14 <i>Use of Socrative & Mentimeter</i> 1. Dr. R. Rajalakshmi, JIPMER,Puducherry <i>Use of Mentimeter to Engage Students During Lecture & Teaching Sessions</i> 2. Dr. Anand Bhasker, CMC Vellore <i>Use of interactive presentation and audience engagement platforms to make lectures and teaching sessions engaging</i> 3. Dr. Arijit Chakraborty, National Sports University, Manipur <i>Chronotype Distribution among University Students in Northeast India: Effects on Sleep Physiology & Metabolic Processes</i>	
1:00-1:30 PM	Valedictory function & Prize distribution			
1:30 PM onwards	Lunch Break			

